Jeehong Kim

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RESEARCH INTEREST

I am interested in designing broadly applicable machine learning techniques for graph-based problems.

- **Keywords**: Graph Representation Learning, Geometric Deep Learning

EDUCATION

Seoul National University, Seoul, South Korea

- M.S.-Ph.D. Integrated Program in Data Science — Mar 2023 – Present
  - Research Interest: Graph Representation Learning
  - Advisor: Prof. Hyunwoo Park
- Planning to fulfill military service as a *Technical Research Personnel* starting around September 2027

Korea University, Seoul, South Korea

- B.S. in Industrial Management Engineering — Mar 2019 – Feb 2023
  - GPA: 3.76 / 4.50

RESEARCH EXPERIENCE

Materials Intelligence Lab, LG AI Research

- Research Intern — Feb 2026 – Present
  - Research Interest: Molecule Property Prediction, Molecular Foundation Model, Geometric Deep Learning
  - Advisor: Dae-Woong Jeong, Sumin Lee, Sung Moon Ko

DSAIL (Data Science & Artificial Intelligence), KAIST

- Undergraduate Research Intern — Jun 2022 – Aug 2022
  - Research Interest: Graph Neural Network, Recommendation System
  - Advisor: Prof. Chanyoung Park

AIDA Lab (Artificial Intelligence & Data Analytics), Korea University

- Undergraduate Research Intern — May 2021 – Jan 2022
  - Research Interest: Anomaly Detection, GAN
  - Advisor: Prof. Sungwon Han

PUBLICATIONS
(†: Equal contribution)

CONFERENCE

[C1] Redefining Maritime Anomaly Detection via Equation-Grounded Synthetic Anomalies
Youngseok Hwang†, Sungho Bae†, Dohun Lee, Jaeeun Seo, **Jeehong Kim**, Wonhee Lee, Hyunwoo Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2026 AI for Science Track (Oral))**

JOURNALS

[J2] STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection
Jeehong Kim†, Minchan Kim†, Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, Hyunwoo Park
IEEE Access (2025.09)

[J1] FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels
Yooyeun Kim†, **Jeehong Kim†**, Wonhee Lee, Hyunwoo Park, Deuk Jae Cho
PLoS One (2025.05)

WORKSHOPS

[W2] Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection
Jeehong Kim†, Youngseok Hwang†, Minchan Kim, Sungho Bae, Hyunwoo Park
NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery

[W1] Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies
Jeehong Kim†, Minchan Kim†, Jaeseong Ju, Youngseok Hwang, Wonhee Lee, Hyunwoo Park
AAAI 2025 Workshop in Anomaly Detection in Scientific Domains

PROJECTS

Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Vessel Behavior Anomaly Detection — Oct 2024 – Dec 2025
  - Anomaly detection using a GNN-based framework on a customized dataset
  - (Published) *Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies*
  - (Published) *STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection*

- Visualization for Ship Collision Risk Assessment — Mar 2024 – May 2025
  - Development of a decision supportive interactive system for ship collision avoidance
  - (Published) *FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels*

AWARDS & SCHOLARSHIPS

Top Reviewer, NeurIPS — 2026

Silver Reviewer, ICML — 2026
- Recognized as silver reviewer (top 50% of reviewers)

Best Poster Award — Dec 2025
- NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery

BK 21 Scholarship, Brain Korea — Mar 2025 – Present

SNU Medical AI Scholarship, Seoul National University — Aug 2024 – Aug 2025

Veritas Scholarship, Korea University — Feb 2022
- Research on detection of anomalies in the manufacturing process
  - Advisor: Prof. Sungwon Han

HAICON2021, Korea Institute of Information Security & Cryptology (KIISC) — Nov 2021
- Construction of an AI model for identifying time-series based anomalies in industrial control systems
  - Ranked 6th out of 177 teams

Industrial Engineering Project Competition, Korean Institute of Industrial Engineers (KIIE) — Oct 2021
- Design of a dynamic corridor network for K-UAM
  - Received participation award in industrial engineering project competition

Data Visualization Competition, Korea University, Yonsei University, Intel — Aug 2021
- Visualization of patient and their deductible premium relationship
  - Received grand prize

PROFESSIONAL SERVICES

Conference Review
- Association for the Advancement of Artificial Intelligence (AAAI) — 2026 – Present
- International Conference on Machine Learning (ICML) — 2026 – Present
- Neural Information Processing Systems (NeurIPS) — 2026 – Present
- Learning on Graphs Conference (LOG) — 2025 – Present

Workshop Review
- AI for Science(AI4Science)@NeurIPS — 2025 – Present

Journal Review
- IEEE Access — 2025

ACTIVITIES

Lab Captain, Data Intelligence and Learning Lab (DIAL) — Dec 2024 – Jan 2026
- Advisor: Prof. Hyunwoo Park

Samsung Dream Class, Samsung Welfare Foundation — Sep 2021 – Feb 2022

Korea-China Leadership Program, Korea Foundation for Advanced Studies (KFAS) — Jul 2019

TEACHING EXPERIENCE

Teaching Assistant
- K-Digital training program at Seoul National University — Jun 2024
- Python data analysis training program for Seoul National University faculty members — Jan 2024

TALKS AND SEMINARS

FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels
- 2024 INFORMS Annual Meeting (Contributed Session) — Oct 2024

REFERENCES

Prof. Hyunwoo Park, Associate Professor, Seoul National University
- Email: hyunwoopark@snu.ac.kr

[*CV compiled on 2026-10-12*]